PRODUCT REQUIREMENTS DOCUMENT

JIGAWA STATE ELECTION SITUATION ROOM (JESR)

Integrated Real-Time Election Monitoring & Reporting Platform

Document Type	Product Requirements Document (PRD)
Product Name	Jigawa State Election Situation Room (JESR)
Version	1.0 - Initial Release
Coverage	Jigawa State, Nigeria - 27 LGAs, 287 Electoral Wards
Platform	Web (Laravel/Livewire), Android & iOS (Flutter), SMS Gateway
Status	Pre-Development - Requirements Finalisation

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1. Executive Summary

The Jigawa State Election Situation Room (JESR) is a purpose-built, multi-platform digital system designed to collect, monitor, analyse, and respond to ward-level election data in real time across Jigawa State. The platform provides a single, auditable source of truth for election-day operations, covering all 27 Local Government Areas (LGAs) and 287 Electoral Wards.

JESR operates across four channels: Web Application, Android App, iOS App, and SMS Gateway. It serves three distinct user layers - Ward Agents (field reporters), Supervisors (LGA coordinators/call desk), and Super Admins (state situation room) - each with precisely defined access rights, responsibilities, and visibility scopes.

The system is engineered for reliability under field conditions: it functions offline with automatic synchronisation when connectivity is restored, accepts minimal SMS-format reports from agents in low-bandwidth zones, and delivers real-time dashboards, escalation alerts, and print-ready reports to state-level decision-makers.

Key Outcomes Addressed by this PRD

- Real-time visibility of voting activities, security incidents, and agent login status across all 287 wards
- Structured, validated data capture with multi-part forms and media evidence attachment
- Three-tier escalation workflows with defined response timelines
- Seven standard report dimensions available for print and export
- Immutable audit trails supporting post-election forensic review and accountability
- Offline-first mobile capability with automatic synchronisation on reconnect

2. Problem Statement & Goals

2.1 Problem Statement

Election monitoring in Jigawa State currently faces critical operational gaps that undermine transparency, responsiveness, and accountability:

- Fragmented reporting: Ward-level data is collected via phone calls, paper forms, or informal messaging channels, resulting in inconsistent, unstructured, and unverifiable information reaching the state situation room.
- Delayed escalations: Security incidents and irregularities are reported late due to the absence of a structured escalation pipeline, reducing effective response capacity.
- No audit trail: Edits, corrections, and decisions made during election day leave no traceable record, making post-election accountability difficult.
- Limited analytics: Without aggregated, filterable data, ward-by-ward, party-by-party, or gender-disaggregated analysis is impossible during or after the election.
- Network unreliability: Field agents in rural wards often lack stable internet connectivity, preventing real-time report submission.

2.2 Product Goals

#	Goal	Success Criteria
G1	Real-time situational awareness	100% of submitted reports visible within 60 seconds across all authorised roles
G2	Structured field data capture	All reports captured via defined multi-part form; validation errors < 5%
G3	Offline-first field operations	Agents can draft and auto-sync reports with zero data loss on reconnect
G4	Rapid incident escalation	Level 1 acknowledged within 10 min; Level 3 within 5 min of flag
G5	Complete audit accountability	Every create/edit/delete action logged with actor, timestamp, and previous value
G6	Inclusive access via SMS	SMS parsing success rate >= 90% for correctly formatted messages
G7	Decision-grade reporting	All 7 standard report dimensions exportable as PDF/CSV within 30 seconds

3. User Personas & Access Model

3.1 Layer 1 - Super Admin (State Situation Room)

Role	State-level election monitoring authority with full system control

Primary Responsibilities:

- Full system control: create/disable accounts, set roles, manage master data (LGAs, Wards, Parties, Senatorial Districts, Candidates)
- Configure system settings: reporting windows, correction windows, SMS templates, push notification channels, and access control rules
- View and export all data; run state-level analytics; generate official reports with signature blocks
- Review all audit logs and perform compliance or forensic review

Visibility: Full access across all LGAs, wards, reports, audit logs, user activities, and SMS flows. All actions recorded with who, when, where, action detail, and device/IP metadata.

3.2 Layer 2 - Supervisor (Call Desk / LGA Coordinator)

Role	LGA-assigned monitoring and validation authority with escalation rights

Primary Responsibilities:

- Monitor incoming reports and messages from assigned LGAs (or statewide if designated)
- Validate, flag, escalate incidents, annotate, or request clarifications from Ward Agents
- Send SMS replies or push notifications to ward agents and stakeholders
- Curate and prepare consolidated LGA reports for Layer 1 review

Visibility: Read access to all reports within assigned LGA scope. Audit log access scoped to assignment. Limited editing rights: append notes, mark verified, mark escalated. Supervisor actions logged with context and justification required for edits or escalations.

3.3 Layer 3 - Ward Agent (Field Reporter)

Role	On-the-ground election observer responsible for ward-level data submission

Primary Responsibilities:

- Submit ward-level reports via mobile app, web form, or SMS using the structured format
- Attach photographic or video evidence as required
- Respond to supervisor queries and confirm corrections within the correction window

Visibility: Own submissions and system alerts/messages only. Cannot view other agents' reports, LGA-level aggregates, or other wards' data. Every submission is logged automatically with timestamp, device ID, and submission method.

3.4 Role & Permission Matrix

Capability	Super Admin	Supervisor	Ward Agent
Create / Manage Accounts	Full	None	None
View All Reports	Full	Scoped (LGA)	Own only
Submit Ward Report	Full	If assigned	Full
Edit Reports	Full	Flag / Annotate	Correction window only
Escalate Incidents	Full	Full	None
Run & Print Reports	Full	Scoped	None
Send SMS	Full	Scoped	Receive only
View Audit Logs	All logs	Scoped logs	Own entries only
System Configuration	Full	None	None
MFA Required	Mandatory	Recommended	Optional

4. Functional Requirements

4.1 Authentication & Access Control

- FR-AUTH-01: Three-role authentication: Super Admin (Layer 1), Supervisor (Layer 2), Ward Agent (Layer 3).
- FR-AUTH-02: Super Admin accounts shall enforce MFA (SMS OTP or authenticator app).
- FR-AUTH-03: Ward Agents authenticate via phone number + password with device binding (phone + device ID).
- FR-AUTH-04: JWT tokens for session management with short expiry and refresh token rotation.
- FR-AUTH-05: Failed login rate-limiting: 5 consecutive failures triggers temporary account lock and Super Admin notification.
- FR-AUTH-06: Row-level data visibility enforced at database query level per role and LGA/Ward assignment.

4.2 Data Capture - Ward Report Form

Part A: Identity & Metadata (Auto-Populated)

- Agent Name: pre-filled from authenticated account; editable only with audit trail entry
- LGA: pre-filled from account profile; dropdown if Super Admin
- Ward: pre-filled from account profile
- Time of Reporting: auto-generated server timestamp; manual override within correction window only, with mandatory audit log entry

Part B: Agents & Security

- Party Registered Agent: searchable dropdown from registered party agents (nullable)
- INEC Agent Name: free-text field (optional)
- Counted Voter Papers: integer, minimum 0
- Security Personnel Present: dropdown Yes / No (required)
- Security Type(s): multi-select, shown only when security_present = Yes | Options: Police, Civil Defence, NSCDC, Army, Vigilante, Other

Part C: Voting Counts

- Time Voting Closed: time picker (nullable)
- Number of Persons Who Cast Votes: integer >= 0
- Voting Papers Left (Not Cast): integer >= 0
- Valid Ballot Papers: integer >= 0
- Invalid Ballot Papers: integer >= 0

Part D: Media, Remarks & Submission

- Photo Attachments: 0-5 photos; accepted formats JPEG, PNG
- Video Attachment: optional; 1 video max; maximum 50 MB
- Audio Note: optional; 1 clip max

- Free-Text Remarks: open text, maximum 1000 characters
- Save as Draft / Submit: correction window applies post-submission (default: 10 minutes, configurable)

4.2.1 Form Validation Rules

- FR-FORM-01: agent_name, lga_id, ward_id, and submission_timestamp are required. Form shall not submit without these.
- FR-FORM-02: All numeric fields accept only non-negative integers. Decimals and negatives are rejected.
- FR-FORM-03: Cross-field check: (valid + invalid) <= number_cast_votes. Violation triggers warning and flags report for supervisor review.
- FR-FORM-04: If security_present = Yes, at least one security type must be selected before submission.
- FR-FORM-05: counted_voter_papers should be >= number_cast_votes (warning only, not a hard block).

4.3 Offline-First & Synchronisation

- FR-OFFLINE-01: Mobile app shall allow draft saving with no internet connection using on-device SQLite storage.
- FR-OFFLINE-02: Offline drafts carry local timestamp (submitted_at_device) and full device metadata.
- FR-OFFLINE-03: On reconnection, app automatically submits pending drafts via POST /api/reports/batch.
- FR-OFFLINE-04: Server assigns authoritative server timestamp but preserves agent's device timestamp for audit.
- FR-OFFLINE-05: Conflicting submissions for the same report ID are flagged for Supervisor review, not silently overwritten.
- FR-OFFLINE-06: Sync status and last sync time are prominently displayed in the agent UI.
- FR-OFFLINE-07: Media uploads queued for background upload; pending media marked with visible "pending media" tag for Supervisors.

4.4 SMS Gateway

Inbound SMS - Report Submission

- FR-SMS-01: Only registered phone numbers are accepted as authenticated SMS senders.
- FR-SMS-02: Required format: JIG REPORT <WardCode> <AgentInitials> <PartyCode> <CountedPapers> <Valid> <Invalid> <Sec:Y/N> <SecCodes>
- FR-SMS-03: Successful parsing creates a ward_reports record (submitted_via = "sms") and sends confirmation: CONF IR <ReportID> Received. Thank you.
- FR-SMS-04: Parse failures trigger an auto-reply instructing sender to correct format or use the mobile app.
- FR-SMS-05: Unregistered phone numbers receive no confirmation; no report is created.

Outbound SMS

- FR-SMS-06: Super Admin and Supervisors can send group broadcasts to defined recipient groups (all agents, by LGA, by ward).
- FR-SMS-07: SMS delivery receipts are stored and displayed in the SMS management interface.
- FR-SMS-08: Failed SMS auto-retried up to 3 times with exponential backoff; after 3 failures the sender is alerted.

4.5 Supervisor Workflow

- FR-SUP-01: Supervisors see a chronological Inbox sorted by severity: escalated > flagged > new.
- FR-SUP-02: Each report exposes quick actions: Verify, Flag (reason required), Escalate, Request Clarification, Call Agent.
- FR-SUP-03: Escalation auto-triggers: status update to "escalated"; SMS to LGA coordinator; email to escalation contacts; audit log entry with actor, timestamp, and reason.
- FR-SUP-04: Supervisors may append notes/annotations within scope. Annotations are appended to audit trail and not editable post-save.
- FR-SUP-05: Request Clarification sends a structured message to the Ward Agent and surfaces the pending query on the agent's screen.

4.6 Audit & Immutable Trail

- FR-AUD-01: Every create, update, and delete action on any report, user, or configuration record generates an immutable audit log entry.
- FR-AUD-02: Audit entries contain: actor user ID, actor role, action type, timestamp (UTC), device/IP metadata, entity type, entity ID, previous value (for updates), and new value.
- FR-AUD-03: Audit logs are append-only. No user, including Super Admin, can delete or modify audit entries.
- FR-AUD-04: Correction window (default 10 minutes) allows agents to amend submitted reports. All corrections retained in audit trail and visible to Supervisors and Super Admins.
- FR-AUD-05: Super Admin can export full audit logs filtered by date, user, entity, or action type.

5. Dashboards & Analytics

5.1 Global Filter Bar (Always Available)

Filter	Options / Input	Applies To
Date / Time Range	Date picker (from / to)	All dashboards
Submission Method	Multi-select: Web, App, SMS	All dashboards
LGA	Dropdown (27 LGAs)	All dashboards
Ward	Dropdown (287 wards, filtered by LGA)	All dashboards
Party	Dropdown (master party list)	Reports, Candidates
Senatorial District	Dropdown	Reports, Candidates
Candidate	Dropdown (filtered by party)	Candidate dashboard
Gender	Multi-select: Male, Female, Other	Agent dashboard
Login Status	Dropdown: Logged In / Not Logged In	Agent status panel
Security Present	Toggle: Yes / No / All	Security incidents panel
Report Status	Multi-select: Draft / Submitted / Verified / Flagged / Escalated	All report views

5.2 Required Dashboard Panels

Panel 1: Agent Login Status

- Filter dropdown: Logged In / Not Logged In; count per LGA and ward
- Drillable list of non-logged-in agents with phone numbers for outreach
- Last login timestamp per agent

Panel 2: Submissions by Ward

- Sortable table: Ward name, LGA, submission count, latest report time, status breakdown
- Click row to drill into individual reports for that ward

Panel 3: Agents by Gender

- Donut chart: Male / Female / Other counts; filterable by LGA and submission status

Panel 4: By Local Government Area

- Choropleth map or sortable list showing LGA-level submission counts, colour-coded by volume and incident severity
- Click LGA to drill into ward-level view

Panel 5: By Political Party

- Stacked bar chart: vote totals and report counts per party; filterable by ward, LGA, senatorial district

Panel 6: By Senatorial District

- Aggregated key metrics: total submissions, flags, escalations, verified reports per district

Panel 7: Candidate Distribution

- Per-candidate reported vote totals; sortable by votes, party, ward, LGA

Panel 8: Security Incidents

- List and map of reports where security_present = true; breakdown by security type
- Severity indicators for escalated reports

Panel 9: Media Gallery

- Grid view of latest media attachments with verification flag overlay
- Sensitive media visible only to Supervisor and Super Admin

Panel 10: Real-Time Feed / Inbox (Supervisor-Facing)

- Live chronological queue of incoming reports and SMS with quick actions (Flag, Escalate, Call, SMS Reply)
- Colour-coded status chips: New (blue), Flagged (yellow), Escalated (red), Verified (green)

5.3 Print & Export

All seven standard report dimensions available as PDF or CSV:

- Agents list by Ward
- Submissions by Gender
- Submissions by Ward
- Submissions by Local Government
- Submissions by Party
- Submissions by Senatorial District
- Candidate Distribution across Wards

All exported reports include: Situation Room name, applied filters, generation timestamp, generator name, and signature block for official print copies. PDF generated server-side with headers, footers, and page numbers.

6. Data Model

6.1 Core Entities

Entity	Key Fields	Description
lgas	id, name, supervisor_ids[]	27 LGAs with assigned supervisor contacts
wards	id, name, lga_id, senatorial_district_id	287 electoral wards mapped to LGAs and senatorial districts
senatorial_districts	id, name	Senatorial zones for aggregated reporting
parties	id, name, acronym, colour_hex	Master political party registry
candidates	id, name, party_id, post_type, ward_id	Candidates per post type and ward/LGA
users	id, full_name, phone, role_id, lga_id, ward_id, gender, party_affiliation, device_id, last_login, is_active	All system users across three roles
ward_reports	id, agent_user_id, lga_id, ward_id, party_agent_id, inec_agent_name, counted_voter_papers, security_present, security_types[], voting_closed_time, number_cast_votes, voting_papers_left, valid_ballot_papers, invalid_ballot_papers, remarks, status, submitted_via, device_id, sync_status, submitted_at_device, created_at	Core election report record
report_media	id, report_id, media_type, storage_url, thumbnail_url, is_sensitive, upload_status, chain_of_custody_notes	Media evidence attached to reports
audit_logs	id, actor_user_id, actor_role, action_type, entity_type, entity_id, previous_value, new_value, reason, ip_address, device_info, created_at	Immutable audit trail (append-only)
sms_messages	id, direction, from_phone, to_phone, body, parsed_report_id, delivery_status, sent_at	SMS transaction log
notifications	id, recipient_user_id, type, message, related_entity_id, is_read, created_at	In-app and push notifications

6.2 Report Status Lifecycle

Status	Description	Who Can Set
draft	Saved locally on agent device; not yet synced	System (auto)
submitted	Successfully received by server	System (auto on receipt)
verified	Supervisor confirmed accuracy	Supervisor, Super Admin
flagged	Marked for attention; reason required	Supervisor, Super Admin
escalated	Elevated incident; triggers notifications	Supervisor, Super Admin

7. API Specification

7.1 Authentication

All API requests (except POST /api/auth/login and the SMS webhook) require a valid Bearer JWT token. Token expiry: 8 hours. Refresh tokens valid: 30 days.

Method	Endpoint	Auth	Description
POST	/api/auth/login	None	Authenticate and return JWT + refresh token
POST	/api/auth/refresh	Refresh token	Obtain new access token
POST	/api/auth/logout	Bearer JWT	Invalidate refresh token

7.2 Ward Reports

Method	Endpoint	Role	Description
POST	/api/reports	Agent / Supervisor / Admin	Submit new ward report
POST	/api/reports/batch	Agent (offline sync)	Batch submit offline-drafted reports
GET	/api/reports/:id	All roles (scoped)	Retrieve single report
GET	/api/reports?lga=&ward=&party=&from=&to=&status=	Supervisor / Admin	Filtered report list
PUT	/api/reports/:id	Agent (window) / Supervisor / Admin	Update report with audit trail
POST	/api/reports/:id/escalate	Supervisor / Admin	Escalate and trigger notifications
POST	/api/reports/:id/verify	Supervisor / Admin	Mark report as verified

7.3 Dashboards, SMS & Audit

Method	Endpoint	Description
GET	/api/dashboard/summary?from=&to=&lga=&ward=	Aggregated state-level summary metrics
GET	/api/dashboard/agents/login-status	Agent login status counts per LGA/ward
GET	/api/dashboard/submissions/by-ward	Submission counts grouped by ward
GET	/api/dashboard/security-incidents	Reports where security_present = true

Method	Endpoint	Description
POST	/api/sms/send	Send outbound SMS to recipients
POST	/api/sms/receive	Inbound SMS webhook from provider (no auth required)
GET	/api/audit?entity=&user_id=&from=&to=	Filtered audit log retrieval (scoped by role)
GET	/api/audit/export	Export full audit log as CSV (Super Admin only)

8. Non-Functional Requirements

8.1 Performance

ID	Requirement	Target
P01	API response time (reads)	< 500ms at 95th percentile under normal load
P02	Report submission acknowledgement	< 1 second for synchronous confirmation
P03	Dashboard render time	< 3 seconds with all active filters applied
P04	Concurrent users	500 concurrent users without degradation
P05	Election day peak load	287 simultaneous agent submissions without failure
P06	PDF export generation	< 30 seconds for standard reports up to 10,000 rows

8.2 Availability

- System uptime target: >= 99.5% during election operation windows (72 hours before through 48 hours after election day)
- System uptime target: >= 99.0% during non-election periods
- Planned maintenance windows scheduled exclusively outside election operation windows
- Automated health monitoring with real-time alerts for API availability, database health, SMS queue backlog, and server resource thresholds

8.3 Security

- All data in transit encrypted using TLS 1.2 or higher; HTTP redirected to HTTPS
- Passwords hashed using bcrypt (minimum cost factor 12) or Argon2
- JWT access tokens expire within 8 hours; refresh tokens rotated on each use
- Media files stored in private S3-compatible object store, served via pre-signed URLs with TTL (default: 1 hour)
- All API endpoints enforce input sanitisation against SQL injection, XSS, and CSRF
- Rate limiting: maximum 10 report submissions per agent per minute; 100 API requests per IP per minute
- Device binding required for Ward Agents; logins from unregistered devices require re-verification
- Penetration testing before go-live and after each major release

8.4 Scalability & Maintainability

- Backend containerised (Docker) and deployable to Kubernetes or managed container services for horizontal scaling
- Database read replicas for dashboard and analytics queries to isolate write operations
- All backend APIs versioned (e.g., /api/v1/)
- Infrastructure monitoring via Prometheus + Grafana configured before go-live
- Centralised log aggregation via ELK stack or equivalent

9. Recommended Technical Stack

Layer	Technology	Justification
Web Frontend	Laravel + Livewire (Blade)	Server-rendered reactive UI; aligned with existing team stack; rapid iteration
Mobile App	Flutter (Android + iOS)	Single cross-platform codebase; native offline SQLite; aligned with team stack
Backend API	Laravel (RESTful)	Unified codebase for web and API; strong ecosystem for auth, queues, WebSockets
Database	PostgreSQL	JSONB for security_types field; row-level security; strong relational integrity
Cache / Real-Time	Redis + Laravel Echo / Pusher	Queue management; real-time WebSocket events for dashboard updates
Media Storage	S3-compatible (AWS S3 / DigitalOcean Spaces)	Scalable object storage with signed URLs and lifecycle cost policies
SMS Gateway	Africa's Talking (primary) / Twilio (fallback)	Strong Nigerian carrier coverage; webhook support; bulk SMS
Authentication	Laravel Sanctum / Passport (OAuth2 + JWT)	Fine-grained token management with MFA addon support
Maps / GIS	Mapbox GL JS or Leaflet.js	Ward/LGA choropleth visualisation and security incident mapping
PDF Generation	wkhtmltopdf or DomPDF (Laravel)	Server-side PDF rendering for official print reports
Monitoring	Prometheus + Grafana	Infrastructure and application performance dashboards
Hosting	Docker + Kubernetes or AWS ECS	Containerised deployment with autoscaling for election day peaks

10. Escalation Matrix & Notifications

10.1 Three-Tier Escalation Framework

Level	Actor	Timeline	Actions Required	Channels
L1	Supervisor	Within 10 min of incident flag	Call agent to confirm facts; update annotation; send SMS to LGA Coordinator	In-app, SMS
L2	LGA Coordinator	Within 30 min	Coordinate with local security actors; prepare consolidated LGA incident report; escalate to Layer 1 if unresolved	SMS, Email, Phone
L3	State / Incident Response	Immediate for violence / major incidents	SMS + Email + Phone to State Security and Steering Committee; prepare public advisory if required	SMS, Email, Phone, Push

10.2 Automated Notification Rules

- Report with security_present = true submitted: Supervisor receives immediate in-app notification and push alert
- Supervisor escalates to Level 2 or 3: system auto-sends SMS and email to all contacts in the predefined escalation list
- Ward Agent report verified or flagged: agent receives in-app and SMS notification
- Clarification request sent to Ward Agent: agent receives in-app alert with question text on their report screen
- System health breach (API latency > 2s, SMS queue > 50 pending): Super Admin and tech support auto-alerted

11. Rollout Plan & Implementation Milestones

11.1 Phased Rollout Strategy

Phase	Scope	Key Deliverables & Acceptance Criteria
Phase 0	Pre-Development	Role/permission matrix; data dictionary; SMS format guide; SOPs; UAT checklist; DB schema seeded with 27 LGAs and 287 wards
Phase 1	Core Build (API + Web)	Auth + roles; report submission API; form Parts A-D; supervisor inbox; audit logging; basic dashboard
Phase 2	Flutter App + SMS Gateway	Offline draft + sync; SMS inbound parser + confirmation; outbound SMS; mobile form parity with web
Phase 3	Analytics + Reports	All 10 dashboard panels; all 7 export report types; PDF print with header and signature; global filter bar
Phase 4 (Pilot)	2 LGAs (1 urban, 1 rural)	User acceptance testing; offline sync validation; SMS parsing load test; supervisor workflow validation; acceptance criteria met
Phase 5 (Waves)	Remaining 25 LGAs in 4 waves (5-7 LGAs each)	Agent enrolment per wave; training delivery; wave-by-wave acceptance sign-off before next wave begins
Phase 6 (Go-Live)	All 287 wards	All 287 agents enrolled and tested; MFA enabled; 24/7 tech support active; escalation contacts published and tested

11.2 Pre-Go-Live Operational Checklist

1. **All 287 Ward Agents enrolled, verified, and completed at least one test submission**
2. All Supervisors assigned to LGAs with confirmed contact details
3. SMS gateway tested: full send/receive round-trip with sample messages from 5 diverse devices
4. Mobile app installed and offline sync tested on representative low-end Android devices
5. Super Admin MFA enabled and credentials secured
6. Escalation contact list published and tested (Level 1, 2, and 3 contacts successfully reached)
7. Backup and restore procedure validated against staging database
8. Monitoring dashboards active: uptime, API latency, SMS queue backlog, database health
9. 24/7 tech support rotation scheduled with clear escalation contacts
10. Security penetration test completed and all critical findings remediated

12. KPIs & Success Metrics

12.1 Operational KPIs

#	KPI	Target	Source
1	% of Ward Agents logged in at least once before election day	>= 95%	users.last_login
2	Avg. time: report submission to Supervisor acknowledgement	< 5 minutes	report timestamps vs audit log
3	% of reports with at least one media attachment	>= 70%	report_media count per report
4	Level 1 escalation response time	100% within 10 min	escalation_log timestamps
5	SMS delivery success rate	>= 95%	sms_messages.delivery_status
6	SMS inbound parsing success rate	>= 90%	parsed_report_id not null
7	System uptime during election window	>= 99.5%	Prometheus / Grafana
8	% ward submissions within 30 min of poll close	>= 80%	voting_closed_time vs submission time

12.2 Data Quality KPIs

- % of submissions failing cross-validation (valid + invalid > cast votes): Target < 5%
- Average edits per report post-submission: Target < 1.2 (indicates training effectiveness)
- Discrepancy rate in random audit sample: Target < 3%
- % of supervisors achieving < 5-minute inbox review time: Target >= 80%

13. Risks & Mitigation

#	Risk	Likelihood	Impact	Mitigation Strategy
R1	Poor network in rural wards prevents submission	High	High	Offline-first mobile app with auto-sync + SMS fallback for critical fields
R2	Agent data entry errors (invalid cross-field totals)	Medium	Medium	Form-level real-time validation warnings; supervisor review of flagged reports; pre-election practice submissions
R3	SMS parsing failures from format deviations	Medium	Low	Clear training on SMS format; auto-reply with format reminder; parser built with common error tolerance
R4	Unauthorised access or data breach	Low	Critical	RBAC + row-level DB security; device binding; MFA for high-privilege users; TLS; pre-go-live penetration testing
R5	Intentionally fabricated reports	Medium	High	Media evidence requirement; random audit sampling; cross-verification; immutable audit trail; phone-binding
R6	System overload during peak election day burst	Medium	High	Load testing before go-live; horizontal autoscaling; read replicas for dashboards; query optimisation
R7	Low agent adoption / form abandonment	Medium	High	Role-based training; quick reference cards; in-app help; pilot with corrections before full rollout
R8	SMS provider outage on election day	Low	High	Dual-provider: Africa's Talking primary, Twilio fallback; automatic failover routing

14. Governance, Training & SOPs

14.1 Steering Committee

A governance steering committee shall be established prior to development kick-off:

- State Chairman (Chair): final authority on policy decisions and public communications
- INEC Liaison Representative: ensures alignment with electoral commission processes
- Security Coordinator: oversees escalation matrix and security actor liaison
- Civil Society Representative: independent oversight and public accountability
- Technical Lead(s): system availability, data integrity, and technical operations
- Communications Lead: public release policy and official summaries

14.2 Standard Operating Procedures (SOPs) to Develop

11. Agent Enrolment & Deactivation: end-to-end process for registering, verifying, and deactivating ward agents
12. Report Submission & Correction Policy: agent workflow, correction window rules, and supervisor review process
13. Escalation Ladder: roles, timelines, communication channels, and decision authority at each level
14. Media Handling & Evidence Preservation: photo requirements, sensitive content tagging, chain-of-custody notes
15. Communications & Public Release Policy: criteria for releasing official summaries, authorisation chain

14.3 Training Programme

Role	Duration	Topics Covered
Ward Agent	90 min	Account setup, form Parts A-D, photo evidence guidelines, SMS format, offline sync, escalation contacts, troubleshooting
Supervisor	2 hours	Inbox workflow, report validation, flagging/escalating, SMS sending, annotation, clarification requests, report export
Super Admin	3 hours	System configuration, user management, dashboard analytics, audit log review, official report generation, data retention

Training deliverables for all roles:

- Quick-reference card: SMS codes, status meanings, escalation contacts
- Step-by-step illustrated PDF guide for form submission and media attachment
- Short video walkthroughs (5-10 minutes per key workflow)
- In-app FAQ accessible from any screen without leaving the app

15. Appendices

Appendix A: SMS Format Reference Card

Full Format	JIG REPORT <WardCode> <AgentInitials> <PartyCode> <CountedPapers> <Valid> <Invalid> <Sec:Y/N> <SecCodes>

Example	JIG REPORT W087 MA APC 120 118 2 Y POLICE,VIG

Token	Example Value	Description
<WardCode>	W087	4-character ward identifier from master ward list
<AgentInitials>	MA	2-3 character initials matching registered account
<PartyCode>	APC	3-letter party acronym from master party list
<CountedPapers>	120	Integer: total counted voter papers
<Valid>	118	Integer: valid ballot papers
<Invalid>	2	Integer: invalid/spoilt ballot papers
<Sec:Y/N>	Y	Y = security present; N = no security personnel
<SecCodes>	POLICE,VIG	Comma-separated: POLICE, NSCDC, ARMY, VIG, OTHER (omit if N)

Appendix B: Field Data Dictionary (Core Report Fields)

Field	Type	Required	Validation Rule	Notes
agent_user_id	Integer	Yes	Must exist in users table	Auto from session
lga_id	Integer	Yes	Must exist in lgas table	Pre-filled from profile
ward_id	Integer	Yes	Must exist in wards; must belong to lga_id	Pre-filled from profile
counted_voter_papers	Integer	Yes	>= 0	---
security_present	Boolean	Yes	true or false	---
security_types	JSON Array	Conditional	Required if security_present = true; canonical list values only	Stored as JSONB
number_cast_votes	Integer	Yes	>= 0	---
valid_ballot_papers	Integer	Yes	>= 0; valid + invalid <= cast_votes	Cross-field check
invalid_ballot_papers	Integer	Yes	>= 0; valid + invalid <= cast_votes	Cross-field check

Field	Type	Required	Validation Rule	Notes
submitted_via	Enum	Auto	web | android | ios | sms	Set by client automatically
status	Enum	Auto	draft | submitted | verified | flagged | escalated	Managed by state machine

Appendix C: Jigawa State LGA Summary

#	LGA	Ward Count	Sample Wards
1	Auyo	9	Auyakayi, Auyo, Ayama, Gafaya, Gamsarka, Gatafa, Gorgor, Kafur, Unik
2	Babura	9	Babura, Burdugau, Dankumbo, Gidan Duna, Gwangwan, Jigawa, Kanya...
3	Biriniwa	9	Biriniwa, Diginsa, Fagi, Gujungu, Machinamari, Matamu, Nguwa...
4	Birnin Kudu	8+	Birnin Kudu, Kanti, Kawaya, Kudai, Lafa, Surko, Wurno, Yalwan Damai...
5	Buji	8+	Amaryawa, Buji, Chamo, Gantsa, Gawar, Kukuma, Kor, Tsangarwa...
6	Dutse	9+	Chamo, Duru, Dustsen Ma, Dustsen Uku, Jigawar Tsada, Kachi...
7	Gagarawa	6	Gagarawa Gari, Gagarawa Tasha, Kore Balatu, Maikilili, Medu, Yalwa
8	Garki	8+	Buduru, Garki, Kanya, Kore, Muku, Raba, Sabon Garin, Wa...
9	Gumel	9	Baikarya, Dan Ama, Garin Gambo, Gumel, Hausawa, Kofar Arewa...
10	Guri	7+	Abunabo, Adiyani, Garbagal, Guri, Kwaila, Margadu, Matara Baba...
11-27	17 remaining LGAs	Varies	Full 287-ward list provided in separate Master Data Seed document

Note: Complete ward data for all 27 LGAs is provided in the Master Data Seed document. The system database must be seeded with the verified 287-ward list before pilot launch.

--- End of Document ---